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Diverse Models, United Goal: A Comprehensive Survey of Ensemble Learning

Ziwei Fan¹ | Zhiwen Yu^{1,2} | Kaixiang Yang¹ | Wuxing Chen^{2,3} | Xiaoqing Liu^{2,3} | Guojie Li¹ | Xianling Yang¹ | C. L. Philip Chen¹

¹School of Computer Science and Engineering, South China University of Technology, Guangzhou, China | ²Pengcheng Laboratory, Shenzhen, China | ³School of Future Technology, South China University of Technology, Guangzhou, China

Correspondence: Zhiwen Yu (zhwyu@scut.edu.cn)

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ABSTRACT

Ensemble learning, a pivotal branch of machine learning, amalgamates multiple base models to enhance the overarching performance of predictive models, capitalising on the diversity and collective wisdom of the ensemble to surpass individual models and mitigate overfitting. In this review, a four-layer research framework is established for the research of ensemble learning, which can offer a comprehensive and structured review of ensemble learning from bottom to top. Firstly, this survey commences by introducing fundamental ensemble learning techniques, including bagging, boosting, and stacking, while also exploring the ensemble's diversity. Then, deep ensemble learning and semi-supervised ensemble learning are studied in detail. Furthermore, the utilisation of ensemble learning techniques to navigate challenging datasets, such as imbalanced and high-dimensional data, is discussed. The application of ensemble learning techniques across various research domains, including healthcare, transportation, finance, manufacturing, and the Internet, is also examined. The survey concludes by discussing challenges intrinsic to ensemble learning.

1 | Introduction

In the realm of machine learning, ensemble learning is a potent method that combines several weak models to produce a robust and formidable predictive model. The fundamental concept behind ensemble learning lies in aggregating predictions from numerous models, often resulting in superior predictive accuracy and enhanced generalisation capabilities compared to individual models. This approach has found successful applications across various domains, including intelligent transportation [1], image processing [2], data mining [3], bioinformatics [4], and medical analysis [5] etc.

Ensemble learning provides notable advantages over individual models, including enhanced predictive performance, superior generalisation capability, adept handling of complex data, and mitigation of overfitting. Ensemble learning overcomes the weaknesses of individual models by using the strengths of numerous models, resulting in more reliable and precise predictions [6]. As depicted in Figure 1, the process can be partitioned into two primary phases: ensemble generation and voting scheme. Initially, given a training dataset, ensemble methods undergo data transformation to yield an array of new training datasets, facilitating the training of an ensemble of classifiers. Upon receiving new data, these classifiers predict labels,

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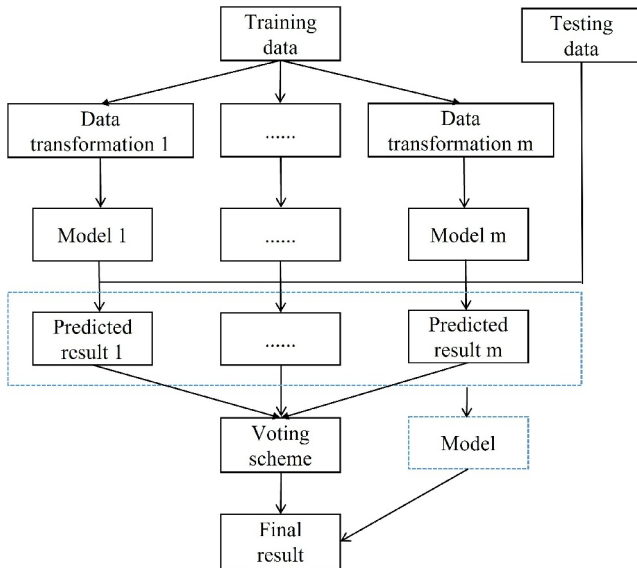


FIGURE 1 | The framework of ensemble learning.

producing a series of label predictions. Subsequently, the appropriate voting scheme, whether majority or weighted, is used to finalise the results.

Ensemble generation techniques encompass various methods for manipulating data, either in the sample space or the feature space. Three categories can be used to categorise ensemble learning approaches based on the various ways of producing individual learners: Bagging approaches, as introduced by Breiman [7], involve creating multiple bootstrap samples from the sample space. Conversely, utilising subsets of features, random subspace techniques work in the feature space [8]. The component models in both the bagging and random subspace approaches operate in parallel, and a majority voting scheme is frequently used to aggregate the final result. Boosting approaches, pioneered by Schapire and Singer [9], focus on adjusting the weights of samples in the sample space. Each subsequent model in boosting optimises the residuals of the previous one. Unlike bagging, boosting models operate sequentially, and a weighted voting scheme is employed to determine the final result. Stacking approaches [10], as depicted in the dashed line of Figure 1, treat the predicted labels from base models as new features. The final ensemble result is then produced by training a new model on these projections.

Ensemble learning has been a research hotspot. In recent review articles, Sagi and Dong et al. [11, 12] summarised representative works in ensemble learning, yet they did not include the latest research achievements. Ganaie et al. [13] conducted a review on deep ensemble learning, while Cao and Mian et al. [14, 15] summarised the applications of ensemble learning in bioinformatics and fault diagnosis. These review papers, however, fall short of covering the overall framework of research on ensemble learning. In this paper, as shown in Figure 2, we provide a four-layer research framework for the research of ensemble learning, which can offer a comprehensive and structured review of ensemble learning from bottom to top. The fundamental methods of ensemble learning, including bagging, boosting, and stacking, will be

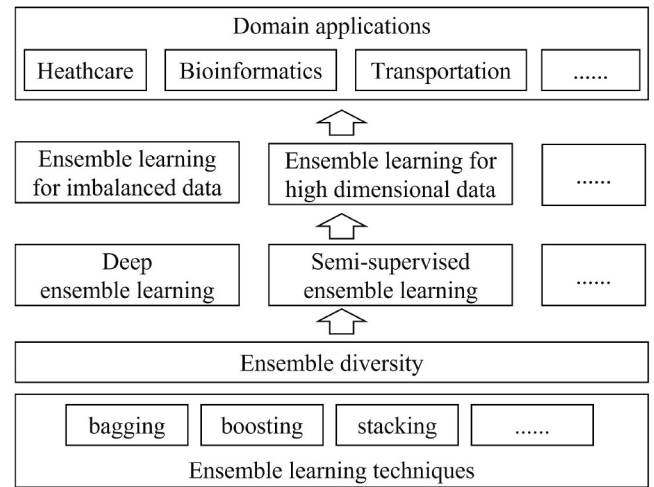


FIGURE 2 | The outline of the survey on ensemble learning.

covered first. The study of ensemble learning diversity is an essential issue for us since the accuracy and variety of base classifiers are the two elements that have the most impact on overall performance in ensemble learning. Therefore, we will subsequently delve into ensemble diversity. Next, we will explore deep ensemble learning and semi-supervised ensemble learning. Because of the complexity of imbalanced data and high-dimensional data, ensemble learning has found extensive applications in these domains. Consequently, we will thoroughly investigate the research advancements of ensemble learning within the contexts of imbalanced data and high-dimensional data. Furthermore, we will discuss the practical application scenarios of ensemble learning, including its applications in fields such as healthcare, bioinformatics, and transportation. Lastly, we present challenges in ensemble learning concerning model generation and voting selection, as well as future research directions. Therefore, this survey endeavours to offer a comprehensive survey of ensemble learning concepts and techniques. We will delve into their strengths and limitations, providing guidelines for the effective application of ensemble techniques in real-world scenarios. To summarise, the following is a list of the contributions made by this review:

- We establish a four-layered research architecture for the field of ensemble learning, including the complete scope of ensemble learning research from basic ensemble learning methods to applications of ensemble learning, and providing a comprehensive and structured review of ensemble learning from the bottom up.
- Based on this four-layered research framework, we summarise the latest research achievements in the field of ensemble learning in recent years and analyse the current research status.
- We identify the current research challenges in ensemble learning related to model generation and voting selection and discuss future research directions.

The remainder of this review is organised as follows: Section 2 describes various ensemble learning techniques, including bagging, boosting, stacking, and ensemble diversity. Section 3

and Section 4 discuss deep ensemble learning and semi-supervised ensemble learning, respectively. Sections 5 and 6 perform reviews on ensemble learning methods for imbalanced data and high-dimensional data. Section 7 provides an overview of the applications of ensemble learning. Section 8 presents the challenges and future directions of ensemble learning. Section 9 draws a conclusion.

2 | Brief Introduction for Ensemble Learning

2.1 | Bagging

Bootstrap aggregating, commonly referred to as bagging, involves training each base model on a different subset after dividing the training data into multiple subgroups. The results of the base model predictions are averaged or voted upon to get the final forecast. The representative algorithms of bagging-based ensemble learning approaches include bagging [7], random subspace [16], random forest [17], and so on. For instance, the random subspace approach trains each base learner exclusively using a random subset of features that are provided as input from the dataset. This allows each learner to focus on different features of the data, improving the model's generalisation ability and effectively reducing the complexity of handling high-dimensional data. The random forest approach also combines the concepts of bagging and decision trees to produce an ensemble of decision trees, each of which is trained with a distinct subset of the training data and a randomly chosen set of attributes. By averaging or voting the forecasts of all trees, the final prediction is achieved.

In recent years, several research studies have further explored bagging-based ensemble learning approaches in terms of model accuracy, model complexity, and model stability. For example, with the same amount of sparsity and complexity, Ditzler et al. [18] extended online feature selection method, which uses bagging and boosting, produced an ensemble model that had a lower error rate than any single linear model. Zhao et al. [19] proposed an ensemble learning-based method using BoostTree as the base learner, the model combines Bootstrap aggregating and boosting to achieve more accurate and reliable performance. To develop depth on the basis of Random Forest and show that deep models may be created without the requirement for backpropagation, a deep model based on nondifferentiable modules was proposed by Zhou et al. [20]. Shaikh et al. [21] proposed a novel sparse ensemble method to improve the robustness of input data under nonstationary distributions. To solve the problems of model sparsity and variety in classification ensembles, Iiduka [22] developed the stochastic fixed-point optimisation approach. The bagging-based ensemble learning models exhibit remarkable performance in terms of stability and prediction accuracy, which has led to their widespread application across multiple domains, such as multi-instance learning [23], multi-view learning [24], and online learning [25]. These applications demonstrate the versatility and adaptability of bagging-based ensemble learning techniques in addressing various challenges in machine learning across different domains.

2.2 | Boosting

Boosting is a sequential ensemble learning method where base models are iteratively trained with each model concentrating on the data that the preceding models incorrectly identified [26]. Figure 3 shows the process of the boosting approach. First, it gives every training example the same amount of weight. The significance of each case is indicated by these weights. Then, the training data is used to train a weak learner (a decision tree, neural network, or support vector machine) with the current weights assigned to the examples. The weak learner is typically trained to minimise the weighted error, where the weights emphasise the importance of misclassified examples. The method then uses the training data to compute the weighted error of the weak learner and increases the weights of instances that were misclassified, making them more significant for the subsequent learner. Correctly categorised instances will have their weights reduced, making them less significant. The algorithm then iterates the process by training more weak learners, repeating steps two and three while updating the weights based on the performance of the prior weak learners. The instances that were incorrectly categorised or given greater weights in earlier iterations will be the focus of the weaker learners that follow. Next, all of the weak learners' predictions are combined to generate the final prediction, usually using weighted voting or weighted averaging. The weights assigned to weak learners may depend on their performance during training. Boosting, to sum up, is an ensemble technique that raises overall accuracy by giving misclassified samples larger weights with each iteration. It consistently constructs a strong classifier by combining weak classifiers.

Table 1 illustrates the popular boosting approaches and their descriptions [27–33, 35]. Recently, Wang et al. [36] designed a technique for online boosting to deal with imbalanced data streams. Masnadi-Shirazi and Vasconcelos [37] studied the cost-

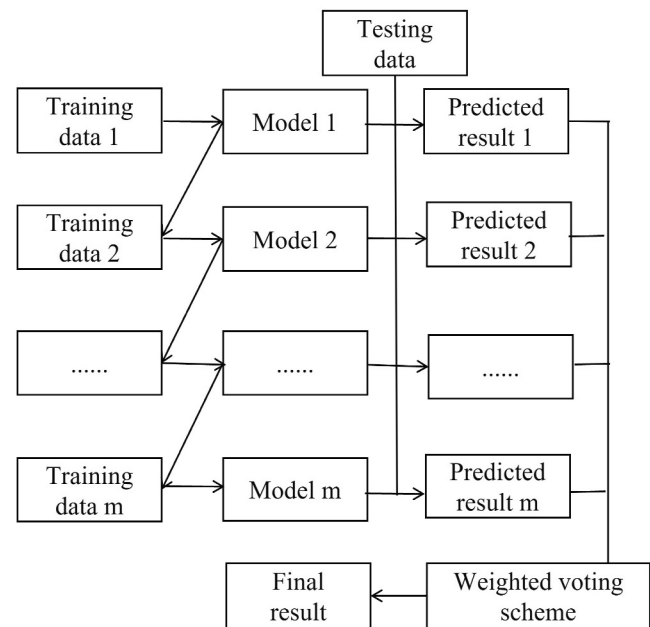


FIGURE 3 | The process of the boosting approach.

TABLE 1 | The introduction of popular boosting approaches.

Methods	The descriptions of the approach	Contribution	Limitation
LogitBoost [27]	It combines weak classifiers by optimising a logistic loss function, and adjusting the weights and the pseudo-residuals of the samples	It specifically targets logistic regression models, and iteratively adjusts them to minimise a logistic loss function	It's sensitive to outliers, and needs well-calibrated weak learners, which can be challenging in some cases
GBDT [28]	It constructs a sequence of decision trees in a sequential manner, with each new tree being trained to fix the errors of the preceding ones	It minimises a predefined loss function, providing high accuracy and efficiency	It is computationally expensive, making it less suitable for real-time applications
LightGBM [29]	A gradient boosting framework that makes use of the unique methods of exclusive feature bundling and gradient-based one-side sampling	It optimises training efficiency, making it particularly suitable for large datasets and high-dimensional feature spaces	It faces challenges in interpretability due to its complex model structure
MART [30]	It combines multiple decision trees by iteratively fitting each tree to the residuals of the previous trees for regression tasks	It excels in regression tasks and provides robustness against overfitting	It is sensitive to outliers and noisy data, and requires careful tuning of hyperparameters to achieve optimal performance
BRT [31]	It adjusts the tree weights to minimise the loss function and produces a strong regression model	It focuses on enhancing regression models by combining weak learners to create a strong predictive model	It is memory-intensive, especially when dealing with large datasets or high-dimensional feature spaces
XGBoost [32]	A flexible end-to-end tree boosting system based on many programming languages is called eXtreme Gradient Boosting (XGBoost)	It incorporates regularisation techniques and parallel processing, making it popular for various machine learning tasks	Its training process is slower compared to other boosting algorithms, and its parameter tuning requires more effort
CatBoost [33]	An open-sourced gradient boosting library for categorical features	It optimises performance while reducing the need for preprocessing categorical variables	Its training process may be slower, and its parameter tuning might require more effort
Multi-class AdaBoost [34]	Without breaking it down into several two-class issues, multi-class AdaBoost simply extends the AdaBoost technique to the multi-class scenario	It combines weak learners to build an ensemble that effectively classifies instances into multiple classes	It's sensitive to noisy data and outliers, potentially affecting its generalisation performance
SemiBoost [35]	An iterative boosting algorithm for semi-supervised learning	It leverages both labelled and unlabelled data to enhance predictive performance	Its performance heavily relies on the quality of the initial weak learner

sensitive boosting approaches. For the regression challenge, Lin et al. [38] investigated the multi-target boosting technique. The deep decision tree transfer boosting technique was created by Jiang et al. [39] by taking into account both the source and target domains. In general, boosting algorithms have been successful in various domains, including classification, regression, and ranking problems. They can significantly improve the predictive accuracy compared to individual weak learners, provide insights into feature importance, and reduce both bias and variance in the model. Boosting algorithms are also renowned for their capacity to manage intricate data linkages and frequently deliver cutting-edge results in both artificial intelligence contests and practical applications.

2.3 | Stacking

By training a meta-model on top of many base models, a technique called stacking, often referred to as stacked generalisation, integrates their predictions [40]. On the basis of the training data, the base models create predictions, which are

then utilised as input features by the meta-model, which eventually learns to make the final prediction.

On the training set of data, stacking initially trains a variety of distinct base models. These models might be of various forms or employ various methods. Then, the trained base models are used to make predictions on the validation or test data that was not used during training. These predictions serve as inputs for the meta-learner. In the following, the algorithm combines the predictions from the base models with the original features, and uses this augmented data to train the meta-learner. The meta-learner learns to create the final predictions using the characteristics from the base model predictions. Once trained, the meta-learner may be used to predict outcomes based on brand-new, unexplored data. The base models generate predictions, which are then fed into the meta-learner to produce the final output. In addition, stacking can be performed with multiple levels, where the base model predictions from one level are used as inputs for the next level. This allows for more complex interactions between the models and can potentially improve performance further.

Recently, there are a lot of advanced stacking methods have been proposed. For example, Jurek et al. [41] explored a novel stacking technique based on cluster analysis. Wu et al. [42] studied the stacking ensemble regression based on Takagi-Sugeno-Kang fuzzy systems. Wang et al. [43] designed a multi-modal stacking approach for real estate appraisal. Chatzimpampas et al. [44] used performance metrics to study stacking ensemble learning. Overall, stacking ensemble learning is a form of meta-learning where the predictions of the base models are used as inputs to another model, called a meta-learner or blender, which generates the final predictions. It is a powerful technique for improving predictive performance by combining the strengths of multiple base models, which has been successfully applied in various domains and can be a valuable tool in machine learning and data science projects, such as prediction of human breast cancer prognosis [45], prediction of therapeutic peptides [46], detection of global navigation satellite system [47], load forecasting [48], bipolar disorder classification [49], and so on.

2.4 | Ensemble Diversity

Ensemble diversity refers to a strategy in ensemble learning where a set of weak models with certain accuracy and diversity are constructed to enhance the overall model's performance. To enhance ensemble diversity, various forms of randomness can be introduced, such as creating different sub-datasets through data sampling, adjusting input feature spaces to generate distinct subspaces, altering output representations, or modifying hyperparameters. This diversity helps reduce the bias of the ensemble model, improving generalisation performance, as the distinctiveness among different weak models allows them to complement each other when handling various aspects of the data or making different types of errors, resulting in more reliable and robust combined predictions.

Table 2 outlines diverse strategies for enhancing ensemble diversity. Firstly, different input data representations or pre-processing methods can be employed to train distinct models. For instance, Yu et al. [50] introduced SSEC, conducting semi-supervised linear classification in disjoint feature subspaces, improving high-dimensional data classification. Secondly, models can initiate training with varied parameters, leading them to different local minima during training; exemplified by You et al. [51] who used multiscale dilated convolution and diverse initialisation for convolutional neural network (CNN)

TABLE 2 | The strategies for the diversity of the ensemble.

Strategies	References
Different input representations	[50]
Different initialisation	[51]
Different learning algorithms	[52, 53]
Data sampling	[54, 55]
Feature sampling	[56]
Injecting noise	[57, 58]
Ensemble pruning	[59–61]

models. Thirdly, employing different learning algorithms in the ensemble captures various aspects of data, as seen in Tahir et al.'s approach [52] to multi-label classification. Guo et al. [53] proposed an ensemble using different base models to address computational challenges in evolutionary optimisation with Gaussian processes. Fourthly, techniques like bootstrapping or random sampling diversify models by training them on different data subsets. Yu et al.'s HCE [54] utilised metric-based data space transformation and adaptive under-sampling for diversity. Their progressive semi-supervised approach [55] used evolving sample selection for diversity. Fifthly, methods such as random subspaces and random forests employ subsets of features for each model. Xu et al. [56] integrated these techniques to enhance ensemble diversity. Sixth, noise injection into data enhances diversity, as shown by Yu et al.'s AP2CE [57] and Melville et al.'s method [58] introducing noise-immune features and artificially constructed training examples, respectively. Seventh, ensemble pruning optimises diversity by selecting suitable base classifiers. Y. Bian et al. [59, 60] used information entropy and neural architecture search for ensemble pruning, while Soto et al. [61] applied static and dynamic pruning to enhance diversity in adaptive boosting (Adaboost) ensembles for various classification problems. Therefore, there are diverse strategies for enhancing ensemble diversity, encompassing methods such as varied data representations, parameter initialisation, learning algorithms, data sampling, feature sampling, noise injection, and ensemble pruning. These techniques address the challenge of achieving diverse and robust predictions in ensemble learning across different application domains, each offering unique ways to bolster the performance of ensemble models.

In summary, the significance of ensemble diversity in enhancing the performance of ensemble methods cannot be overstated. Striking the right balance is essential, as an excessive level of diversity can result in an ensemble of weak models prone to making numerous mistakes, while insufficient diversity may lead to an ensemble that fails to outperform its individual constituents. Achieving an optimal level of diversity is key to reaping the benefits of ensemble learning techniques across various applications and problem domains.

3 | Deep Ensemble Learning

Deep ensemble learning combines the strength of deep neural networks with the advantages of ensemble methods to improve performance, robustness, and generalisability. In this section, we will go into the specifics of training methodologies and highlight how important feature fusion is to maximise the effectiveness of deep ensemble techniques.

3.1 | Bagging for Deep Ensemble Learning

Deep ensemble learning during the training stage is the use of different ensemble strategies to integrate deep models with different structures or parameters, thereby enhancing the overall performance and robustness. Bagging, also known as bootstrap aggregation, is used to improve deep base classifiers.

The idea is to create a set predictor based on a series of observations better than a single prediction made from raw data. The resulting independent observations match the original data in terms of size and distribution. Adding bagging to the original model required two steps: generating bagged samples, passing them to the base model, and combining multiple prediction factors. It is possible to get bagged samples with or without replacements. The output of the combined basic predictor changes since most majority votes are used for classification, and averaging is used for regression.

The most direct approach to integrating multiple deep neural networks is through direct integration. For example, Khwaja et al. [62] utilised bagged neural networks for prediction tasks, achieving improved performance by merely averaging the output from each trained network. Simultaneously, in a cost-sensitive selected deep ensemble, the Group Method of Data Handling (GMDH) technique—a data-driven modelling strategy for model selection and optimisation—is used to enhance the final prediction outcomes. Ensemble learning can be classified as homogeneous or heterogeneous depending on whether the ensemble models have the same structure. Homogeneous ensemble consists of the same type of base learners, aiming to enhance model stability. A heterogeneous ensemble blends various learner types in an effort to maximise diversity. Routt et al. [63] created an open-source, highly accurate deep ensemble learning-based pipeline for classifying the morphology of red blood cells (RBCs). Furthermore, RBC morphology is classified by a deep ensemble of four CNNs employing adaptive thresholding or semantic segmentation to get an unweighted averaging of models. Pneumoconiosis disease was identified in chest X-ray radiographs (CXRs) by Devnath et al. [64] using a variety of deep neural network models. Three ensemble learning strategies (simple averaging, multi-weighted averaging, and majority voting) were employed to assess performance on randomised cross-fold and leave-one-out cross-validation datasets. Chakladar et al. [65] proposed an ensemble Electroencephalography (EEG) classification model that uses multiple concurrently operating, similar structured LSTM networks. The ensemble model (i.e., a user's cognitive state) is computed using the average weighted combination of individual model predictions. Egele et al. [66] proposed an automated method of creating ensembles of deep neural networks using joint neural architectures and hyperparameter searches. The law of total variance is employed to break down the prediction variance of deep ensembles into two categories: uncertainty related to the aleatoric (data) and epistemic (model). The phased optimisation of deep ensemble models is also a widely adopted optimisation method. The optimisation process is systematically divided into distinct phases, each of which focuses on specific aspects or components of the system, aiming to enhance efficiency and convergence in the optimisation process. As a deep ensemble dynamic learning network, Zhang et al. [67] developed a two-stage bagging dynamic learning network that is based on neural dynamic learning and bagging algorithms. By classifying extracted feature samples by pre-trained models, the models successfully diagnosed pneumonia and its type. As part of their heterogeneous ensemble classifier approach, Das et al. [68] proposed a biomedical image classification method in which four heterogeneous base classifiers, including CNN, RNN, LSTM, and GRU, are considered, metadata is generated, and a

fuzzy min–max model is employed to reduce uncertainty. Sohail et al. [69] assessed mitotic nuclei in breast histopathology images by combining five distinct architecture deep CNNs with multilayer-perceptron meta-classifiers. The resulting classifier is precise and robust, providing a reliable final decision. Wang et al. [70] developed a deep ensemble fuzzy classifier for imbalanced data classification tasks with high interpretability and promising classification performance. A deep ensemble of Takagi-Sugeno-Kang (TSK) fuzzy subclassifications takes advantage of the minority class and problematic areas for high classification performance and interpretation.

Overall, Bagging in deep ensemble learning offers advantages such as reducing overfitting, improving generalisation performance, and simplifying model selection. However, it also has drawbacks, including higher computational costs, sensitivity to the dataset, and so on. In real-world applications, the particular problem and dataset properties should be taken into consideration while selecting an effective ensemble learning method.

3.2 | Boosting for Deep Ensemble Learning

By using boosting technology, we can improve the generalisation of weak learning models. When it comes to classification tasks, linear combinations of weak learners or majority voting produce results that are superior to those of a single weak learner. AdaBoost and Gradient Boosting are two well-liked boosting techniques.

By adding appropriately weighted predictors at every iteration, AdaBoost minimises misclassification losses. The deep boosting model [71] builds an ensemble model by employing deep decision trees. It can be used in conjunction with any other rich family classifier to enhance generalisation performance. Deep boosting consists of multiple steps, where the decision of which classifier to add and the weights to apply depends on the classifier's (data-dependent) complexity. Interpreting the deep boosting classifier using the concepts of structural risk reduction is a crucial part of every learning stage. Han et al. [72] used IBCNN to prevent overfitting by accumulating information from multiple training batches. A decision stump on top of a single neuron serves as a weak learner, and each mini-batch of weights is learnt using the AdaBoost method. By combining boosted deep belief networks (DBN) [73] with multiple DBNs, we are able to generate a high-performing facial expression recognition system. Using an iterative approach, the model builds a strong classifier using complex feature representations. Instead of the image patches used by DBN, IBCNN [72] trains the weak classifiers using the whole image. Loss functions from weak learners and the global loss function are combined in the IBCNN model to improve its efficiency.

Boosting was employed by Moghimi et al. [74] to train deep CNNs. Rather than averaging, CNN was designed to incorporate the boosting weights using the least squares objective function. Additionally, they showed how network structure can take the place of CNN in their boosting architecture to enhance the performance of basic classifiers. The concept of dense connections was developed by a deep boosting framework [75] to address the issue of vanishing gradients for image denoising,

which raises the complexity of training the networks owing to boosting. Using a dilated dense fusion network, Chen et al. [76] utilised the deep boosting architecture to boost the performance of image restoration. Yang et al. [77] used CNNs to analyse rolling bearing vibration signals and extract fault-related preliminary features. Using both Gradient Boosting Decision Trees (GBDT) and nonequivalent Cost Logistic Regression (NCLR), the extracted features are optimised by quantifying their importance, contributing to rolling bearing faults, for fault classifications with optimised nonequivalent costs assigned to fault severity. Theoretical convergence guarantees for online boosting were demonstrated by Beygelzimer et al. [78]. Increased convergence guarantees are demonstrated by batch-boosting techniques that use online boosting.

In summary, boosting for deep ensemble learning effectively addresses the vanishing gradient problem, enhances model generalisation, and achieves improved performance on complex tasks. However, its training process can be computationally intensive, and the sequential nature may lead to longer training times. Additionally, overfitting can be a concern in certain scenarios. Further research is needed to optimise its computational efficiency and mitigate potential overfitting issues.

3.3 | Stacking for Deep Ensemble Learning

Stacking is an ensemble method that combines the results of several base models by using the meta-learning model. It can be accomplished by choosing the 'best' base model or by in some way merging the results of several base models.

A two-step optimisation technique was presented by Zhao et al. [79] to construct a deep ensemble model based on weighted voting that can be applied to classification challenges. In the first phase of creating the ensemble model, a bi-objective algorithm was used to generate a set of candidates for classifiers, which required taking the diversity of the classifiers and the accuracy into account. After selecting representative classifiers, the next step was to determine the weights of the classifiers. An alternative was added to the second stage of the procedure in the form of a tree-based nonrepetitive evaluation mechanism. Deng et al. [80] presented a deep stacking network (DSN) as a technique for parallel training on huge datasets. By employing the kernel method, the number of hidden layers in Deng et al. [81]'s kernelised DCN, or kernel deep convex networks, neared infinite. Based on kernelised DCN results, K-DCN performed better than DCN. Using a combination of model-based optimisation and hill-climbing techniques, Welchowski and Schmid [82] offered a framework for parameter estimation and model selection in kernel DSNs. Like sparse coding, deep learning uses another popular technique called sparse coding. Sparse representation has various advantages, such as noise robustness, effectiveness for learning useful features etc. Sun et al. [83] proposed an image-classifying algorithm called Sparse S-DSN is proposed for the detection of abnormalities in images. The convex optimisation algorithm and gradient descent algorithm were used to solve weights in Sun et al. [83] sparse simplified neural network modules. Zhang et al. [84] stated that a multi-patch hierarchical network can deblur images by stacking

multiple patches. In order to enhance the neural architecture search performance, Wang et al. [85] repeatedly stacked blocks.

Data-driven frameworks were used by Welchowski et al. [76] to estimate parameters, tune hyperparameters, and select models for kernel DSN. Li et al. [86] developed sparse simplified neural network modules that learn the local dependencies between hidden units by grouping the hidden units or representations. DSN is also applied in Deep Reinforcement Learning. By merging data from two formal networks, the Grasp network and the Stacking network, Zhang et al. [87] developed an integrated robotic arm system based on the DSN approach. In R-DSNs [88], the advantage of DSNs is combined with the advantage of recurrent neural network (RNN). Zhou et al. [89] stacked extreme learning machines were presented. The stacking concept has also been applied to traditional models, such as Random Forests, to extend deep architectures or deep forests [20]. According to Kang et al. [90], their model produced notable performance increases with fewer parameters and training samples by stacking conditionally restricted Boltzmann machines and deep neural networks. Low et al. [91] provided a stacking-based deep neural network trained without backpropagation.

Stacking for deep ensemble learning leverages diverse base models, leading to enhanced generalisation and performance across various tasks. However, its effectiveness heavily relies on the selection of suitable base models, and the training of multiple models can demand substantial computational resources. Additionally, stacking may become less beneficial when base models are similar, and the risk of overfitting should be carefully managed. Further investigations are required to refine the selection process for base models and optimise computational efficiency.

3.4 | Optimising Deep Ensemble Models

To optimise deep ensemble models, adversarial learning is an effective training method, which introduces adversarial examples into the training data to enhance robustness. For example, Li et al. [92] introduced an adversarial training technique that perturbs a malware example without compromising its malicious functionality by utilising several generation approaches and manipulation sets. Ahmadian et al. [93] also presented an ensemble approach using reinforcement learning to combine the three similarities of data features into a similarity value to boost recommender systems, where the data features are based on stacked denoising autoencoders. Using deep ensemble reinforcement learning, Ghosh et al. [94] created a deep ensemble multi-agent training technique that can arbitrate between local and more expansive models. With the use of ensemble learning, models' robustness and generalisation are enhanced, facilitating improved recognition and decision-making in out-of-distribution scenarios. The incremental signal classification (ISC) technique identifies and categorises unknown signals, basically an open-set classification task. In the study by Yang et al. [77], a Deep Ensemble Siamese Network was developed to detect unseen categories and the incremental accumulation of signals from the discovered classes. Once the

accumulated samples have been gathered, the OCN is updated with the new signals for incremental learning. A straightforward yet efficient technique for lowering picture noise was presented by Liu et al. [95] using Bayesian deep ensemble for real-world image denoising. To improve resilience in real-world noisy images by modelling prediction variance, a Bayesian deep learning technique was devised, which predicts weighting uncertainty and weighting maps instead of learning pixel-by-pixel weighting maps. A hybrid ensemble classifier was proposed by Parveen et al. [96] employing fuzzy logic, a DBN, and a neural network. The recovered characteristics are used to train a neural network, and the knowledge bases that are obtained from the neural network are used to teach fuzzy logic.

Some studies focus on designing efficient constraints, for example, a study by Xiao et al. [97], which suggested a GMDH-based selected deep ensemble model employing the GMDH, focuses on setting effective constraints. The CNN-RNN, CNN-LSTM, and CNN-GRU are built in this model using CNN, RNN, LSTM, and GRU, in that order. In addition to CNN, four focused loss-based deep neural networks are constructed in order to generate the fundamental forecasts. The deep ensemble neural-like (DEN) P system was proposed by Xue et al. [98], which is also a parallel neural-like P system. Its foundation is the fusion of deep CNNs with spiking neural P systems (SN P systems) to enhance the precision and effectiveness of separating CSC lesions and ellipsoid zones from the Bruch membrane (EZ-BM) regions. Wu et al. [99] developed an ensemble pruning approach based on hierarchical ensembles that can identify smaller, high-quality deep ensembles of size $S \ll M$, or more miniature, high-quality deep costumes for a given pool of M base models that are better than the total ensemble of all M models. They developed a focal diversity consensus technique to identify smaller, deep ensembles with low correlations, based on a focal diversity-based hierarchical pruning algorithm.

3.5 | Feature Fusion of Deep Ensemble Methods

Deep ensemble learning has been extensively studied, and a critical dimension within this field is gaining prominence and contributing significantly to its effectiveness—feature fusion. Feature fusion in deep ensemble learning is pivotal in combining information from multiple sources, each offering a unique perspective on the underlying data. This technique extends beyond traditional ensemble methods that merely aggregate predictions and focuses on harmonising the diverse feature representations learnt by individual base models.

Asif et al. [100] proposed a framework combining body, face, and hand features for video-based action recognition. It encodes more efficient spatiotemporal characteristics by encoding temporal and spatial features that are resilient to noise in posture estimation. The deep action stamps provide it with exceptionally accurate convolutional and structural feature learning. As an alternative to the traditional approach to HSI classification, Dong et al. [101] developed an ensemble CNN method for HSI classification with better generalisation ability. As a result,

classifiers can get discriminant spectral-spatial features by extracting and fusing spatial information with original spectral bands. Senousy et al. [102] introduced Multi-level Context and Uncertainty-aware, a dynamic deep learning ensemble model. It is a combination of several context-aware, multi-level models. It is a layer-wise way to learn how image patches are related, so the model can use an uncertainty quantification component to do a dynamic ensemble model that is incredibly sensitive to contextual information. In order to capture the temporal dynamics throughout the sequence, temporal dynamics along the sequence, and distinct characteristics in the temporal domain, Navienkumar et al. [103] proposed three subnets called SNet, TNet, and BodyNet. A deep ensemble network SEE-Net [104], that learns to express spectral self-expressive properties, has been used to form a single band of hyperspectral data. This shows how one round can help develop hyperspectral data. The frames were split into three-channel false-colour images to extract deep features from hyperspectral videos. By optimising the knowledge distillation elements as hyperparameters, Okamoto et al. [105] developed a knowledge distillation design for the ideal ensemble based on maximising ensemble accuracy through graph structure optimisation. A deep ensemble model for diagnosing bearing faults was proposed by Chen et al. [106] based on residual networks and bidirectional long short-term memories (BiLSTM). With the help of an ensemble learning multi-branch model, associative features were extracted from the original 1-D time-series signals. Using three well-performed pre-trained models, including VGG16, ResNet152V2, and DenseNet201, Singh et al. [107] presented a deep ensemble medical image captioning network. Li et al. [108] talked about the trade-off between accuracy and deadlines. Deep ensemble models now have a query difficulty-dependent task scheduling framework, which reduces deadline misses by five and improves accuracy by 30.8% in latency-sensitive tasks compared to conventional ensemble models. An effortless ensemble was proposed by Hasegawa and Kondo [109] for HAR, which facilitates deep ensemble learning within the same model. Additionally, they gave EE many techniques, including input variations, sequential ensemble, and channel shuffle.

In summary, deep ensemble learning successfully integrates the various feature representations that individual models have learnt, so combining the best aspects of both deep learning and ensemble learning. This method has a lot of potential for handling complex problems and can lead to significant gains in accuracy, generalisation, and flexibility in a variety of fields.

4 | Semi-Supervised Ensemble Learning

Semi-supervised ensemble learning is an amalgamation of semi-supervised and ensemble learning techniques. Models' capacity for generalisation is improved through the use of labels and unlabelled data in semi-supervised learning [110, 111]. Compared to fully supervised learning algorithms that employ only a small quantity of labelled data, it has been demonstrated that using unlabelled data during the training process can result in superior performance.

4.1 | The Process of Semi-Supervised Ensemble Learning

As shown in Figure 4, the process of semi-supervised ensemble learning typically involves three steps: generation of base learners, utilisation of unlabelled data, and combination of base learners. A semi-supervised ensemble learning approach starts with the generation of numerous base learners, which can be support vector machines, decision trees, or neural networks. The labelled data are used to train these base learners through conventional supervised learning techniques. Each base learner produces its own predictions based on the labelled data. For instance, a model ensemble may include a conditional random field model, bidirectional gated recurrent units, and BiLSTM [112].

Then, the algorithm utilises the unlabelled data to enhance the ensemble model. Various techniques can be employed, such as self-training [113], co-training [114], transductive learning [115], and so on. It is usually done by training a model on a small set of labelled data and then predicting labels for unlabelled data using that model. A model is retrained iteratively using the most confident predictions. The co-training process involves training two or more base learners on different views of the labelled data. Unlabelled data are first predicted to train the ensemble learners, with the best predictions added to the training set. The process is repeated iteratively. Transductive learning aims to directly predict labels for unlabelled data without explicitly learning a model. Labels are frequently propagated from labelled instances to neighbouring unlabelled ones based on similarities or differences between the two instances. These techniques use unlabelled data to refine or generate predictions. With ensemble semi-supervised gated stacked autoencoders, you can train different submodels from different unlabelled datasets to ensure their diversity [116].

The final step in semi-supervised ensemble learning is combining the predictions from the labelled and pseudo-labelled instances to form the final ensemble prediction.

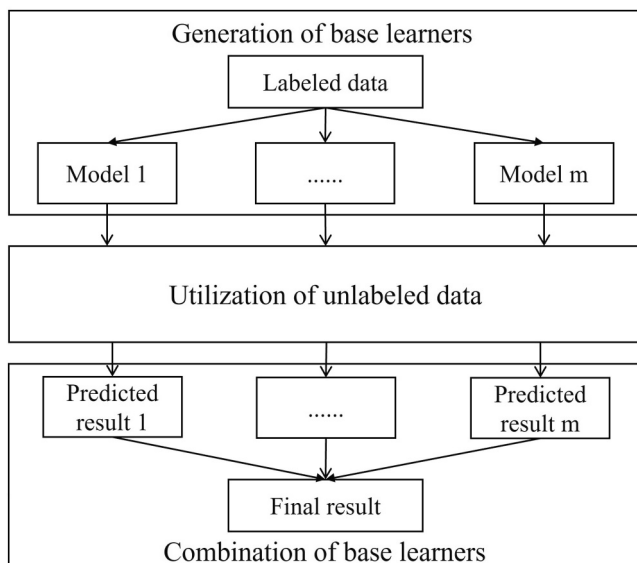


FIGURE 4 | Three stages of semi-supervised ensemble training.

Common aggregation techniques include majority voting, weighted voting, and averaging. Predictions can be improved by integrating a classifier with a weighted voting scheme based on classification accuracy [117]. To enhance the ensemble model's overall performance, it makes use of each learner's unique strengths and minimises their shortcomings.

4.2 | The Advantages of Semi-Supervised Ensemble Learning

When compared with traditional ensemble learning approaches, semi-supervised ensemble learning [1,8–121] offers several advantages as shown in Table 3. First, unlabelled data provides additional information and can help in discovering hidden patterns or structures in the data. The use of labelled data can be of particular benefit when the availability and cost of labelled data are limited. For instance, Dupre et al. [122] showed that by introducing unlabelled data into the training process of semi-supervised ensemble learning, it is feasible to increase the accuracy and generalisation capabilities of models. Karim et al. [123] proposed a new semi-supervised learning method which employs a uniform selection mechanism to combat high label noise and improve generalisation performance. This method leverages unlabelled data to enhance the quality of pseudo-labels and demonstrates the potential of semi-supervised learning in handling label noise effectively. Second, semi-supervised ensemble learning can be computationally efficient due to the iterative boosting algorithm used in the training process. For example, Xiao et al. [124] developed a computationally efficient semi-supervised ensemble method for intrusion detection that iteratively trains base classifiers on pseudo-labelled data, avoiding expensive manual labelling efforts. Moreover, by utilising the unlabelled data, it reduces the need for extensive labelling efforts and can make more efficient use of available resources. Krasakis et al. [125] showed that their self-training based semi-supervised ensemble approach achieves strong performance with minimal labelled data, reducing annotation costs. Third, by incorporating unlabelled data, semi-supervised ensemble learning can enhance the diversity among the base learners. For example, Yang et al. [126] described a class-aware contrastive semi-supervised learning approach that handles in-distribution and out-of-distribution data separately to enhance the robustness of the model. This approach promotes diversity among base learners, leading to improved predictive performance and robustness. Fourth, semi-supervised ensemble learning provides flexibility in incorporating prior knowledge related to the weight of the base learners. For instance, Paenroodposhti et al. [127] presented a semi-supervised framework that uses multi-task learning and logic constraints to integrate past knowledge into deep neural networks. This allows encoding domain knowledge to guide the learning process and improve model performance. Fifth, semi-supervised ensemble techniques such as self-training and pseudo-labelling iteratively re-train the models, weeding out poor base learners that cannot improve. The ensemble approach allows for the combination of multiple base learners, mitigating the impact of individual weak models and improving overall performance. The semi-supervised ensemble framework proposed by Li et al. [128], which was introduced for learning with

TABLE 3 | The advantages of semi-supervised ensemble learning.

Advantages	References
Performance improvement and robustness to label noise	[122, 123]
Efficient computation and cost-effectiveness	[124, 125]
Enhanced diversity among base learners	[126]
Flexibility in incorporating prior knowledge	[127]
Robustness to poor base models	[128, 129]

noisy labels, is a great example. This framework robustly trains graph neural networks and performs label correction simultaneously, showcasing the robustness of semi-supervised ensemble learning against poor base models and label noise. Kong et al. [129] introduced a dependable semi-supervised ensemble learning technique that uses unlabelled data to produce a variety of classifiers via self-training. After that, an ensemble of these classifiers is assembled for prediction purposes.

Overall, semi-supervised ensemble learning is a promising research direction that combines the strengths of ensemble learning and semi-supervised learning. It is especially helpful in situations when labelled data is scarce or expensive to gather because it provides a means of utilising vast volumes of unlabelled data to enhance the performance and generalisation of ensemble models.

5 | Ensemble Learning for Imbalanced Data

The majority of existing classification algorithms make the assumption that the data used for training have a balanced class distribution, meaning that each class has roughly the same amount of samples. When dealing with imbalanced class data, conventional classification algorithms that primarily focus on achieving high overall classification accuracy tend to disproportionately emphasise the majority class, resulting in a degradation of classification performance for minority class samples [130]. With the development of ensemble learning techniques, more and more research has introduced ensemble learning to the problem of class imbalanced data. In essence, the class imbalance issue is addressed by ensemble methods, which entail the creation of multiple distinct predictive models and subsequent fusion of their predictions, typically achieved through methods such as averaging or voting.

5.1 | Bagging for Imbalanced Data

Bootstrap sampling techniques are incorporated into the aggregation process known as bagging. By averaging or voting the predictions made by several learners on various samples from the same dataset, it increases the model's accuracy. This helps in solving the bias-variance trade-off problem while avoiding overfitting and ignoring correlated features between samples. Bagging is suitable for homogeneous weak learners and allows

them to learn and cooperate in parallel, thus improving classification performance.

In recent years, many new strategies have been proposed to augment traditional Bagging algorithms to deal with the imbalanced data problem. For example, Jerzy et al. [131] presented a novel method called Neighbourhood Balanced Bagging. The method takes into account the local characteristics of the minority class distribution and explains how the performance of ensembles on imbalanced data varies depending on the minority examples in the neighbourhood. Wang et al. [132] proposed ensemble-based algorithms that combine various resampling methods and introduced a novel approach called SMOTEBagging to address multi-class imbalanced datasets. Using random subspace SVM and asymmetric bagging, Tao et al. [133] enhanced relevance feedback performance in content-based picture retrieval. They did this by solving concerns with small-sized training sets, class imbalance, and large feature dimensions. In order to overcome class imbalance and improve classifier diversity, Sun et al. [134] introduced the DTE-SBD, a unique Decision Tree Ensemble model that employs synthetic minority over-sampling (SMOTE) and bagging with differential sample rates. Meanwhile, a large number of deep learning methods have been introduced into ensemble learning to enhance the performance of learning from imbalanced data. In order to overcome the class-imbalanced learning problem, Chen et al. [135] incorporated ensemble learning into deep CNNs. An ensemble of auxiliary classifiers branching out from different hidden layers of a CNN is trained end-to-end with the CNN. In Wu et al.'s [136] approach, the complementarity of feature learning and model diversity are improved by using both an ensemble-learning-based CNN and a 1D-CNN model.

Combining Bagging with data preprocessing methods is frequently more straightforward and effective than utilising boosting ensembles. Bagging algorithms obviate the need for recalculating various weights, thereby eliminating the necessity to fine-tune weight update formulas or modify algorithmic computations. The pivotal consideration in these approaches revolves around effectively addressing class imbalance while simultaneously upholding the significance of diversity.

5.2 | Boosting for Imbalanced Data

The core idea of Boosting in dealing with imbalanced data classification is to train multiple weak learners sequentially and give more attention to misclassified instances in each subsequent learner. This adaptation process helps to better model a small number of classes, reducing bias and improving overall prediction accuracy. This is because they focus on learning from challenging instances and adjusting the weights throughout the training process. This approach allows boosting algorithms to capture complex relationships in imbalanced datasets.

In recent years researchers have developed new Boosting methods to handle the acute imbalanced data problem. For example, Anna et al. [33] proposed the CatBoost algorithm to efficiently handle classification features, with superior performance on a variety of datasets and faster implementation on

GPUs and CPUs compared to existing gradient boosting libraries. Using image processing and machine learning methods, Bartosz et al. [137] presented an automated clinical decision support system for the malignancy grading of breast cancer. With the help of the EUSBoost ensemble classifier, which performed very well on a real, unbalanced medical dataset, they were able to successfully address the imbalanced problem. In order to secure genuine and structurally dispersed majority samples throughout the undersampling process, Wang et al. [138] introduced a novel entropy and confidence-based undersampling boosting framework that addresses unbalanced situations. The framework uses both entropy and confidence as benchmarks. Aboozar et al. [139] introduced AdaBoost-CNN, a machine learning method that combines AdaBoost with CNNs to effectively handle large imbalanced datasets, achieving higher accuracy and computational efficiency compared to traditional AdaBoost and baseline CNN methods.

Ke et al. [29] introduced LightGBM, a novel GBDT implementation that significantly improves efficiency and scalability by utilising GOSS and EFB techniques, achieving up to over 20 times faster training compared to conventional GBDT with almost the same accuracy on various datasets. A distinctive feature of LightGBM is that it employs a histogram-based technique for constructing decision trees, which randomly selects a subset of the data, thereby increasing efficiency and preventing overfitting, especially in cases with a large number of features. When it comes to imbalanced datasets, LightGBM offers a variety of features and techniques [140, 141], including the flexibility to employ diverse objective functions and assign distinct weights to various classes. This capability to address imbalanced datasets renders LightGBM a compelling choice for a broad spectrum of applications.

Boosting algorithms are able to adaptively regulate the trade-off between model complexity and accuracy, as well as the learning rate and number of enhancement iterations. Although lifting is effective, it may still be sensitive to noisy or anomalous data points. Additionally, because of its sequential structure, it could be computationally costly, particularly for big datasets. Nonetheless, boosting remains a prominent option for addressing model performance in imbalanced data and augmented reality scenarios.

5.3 | Cost-Sensitive Ensemble for Imbalanced Data

When the classes are not evenly distributed, the generic ensemble learning Adaboost method employs accuracy as the classification assessment criteria, and this tactic favours the majority of the classes because it contributes more to the total accuracy. In order to solve this problem, numerous research [142–144] have explored alternative weight updating strategies for AdaBoost, aiming to give more emphasis to the classes with fewer samples during the training process. To establish this discrepancy in imbalanced data, cost-sensitive methods maintain the fundamental learning structure of AdaBoost. Concurrently, they integrate cost sensitivity into the weight adjustment equation.

These methods usually differ in the way they modify the weight update rules. The most representative algorithms include CSB1, CSB2 [142], AdaCost [143], AdaC1, AdaC2 and AdaC3 [144]. The cost adjustment function is added to the weight updates in the AdaCost algorithm. This function changes the weight for samples with greater cost weights if the sample is incorrectly categorised and reduces it otherwise. Sun et al. [144] addressed the challenge of imbalanced class distribution in data classification, highlighting the need for specialised research efforts, and explores cost-sensitive boosting algorithms built upon the AdaBoost meta-technique to improve classification accuracy, particularly in medical datasets characterised by class imbalance. AdaCost mainly makes two improvements on the basis of Adaboost: (1) Assigns a misclassification cost factor to each training sample and adds the cost factor to the learning process of classification. (2) Adds a cost adjustment function to the weight adjustment formula, so that samples with high misclassification cost have relatively high weights and samples with low misclassification cost have relatively low weights. Nevertheless, these cost-sensitive ensemble methods still retain the sequential structure of AdaBoost, which implies a decrease in learning efficiency.

Recently there have been an increasing number of novel weighted base classifiers integrating learning methods to cope with the imbalanced classification problem. For example, Yang et al. [145] proposed a weighted broad learning system (BLS) to solve problems related to imbalanced data, outliers, and noise, which includes adaptive weighted BLS (AWBLS) and incremental weighted ensemble BLS (IWEB), and a large number of experiments have shown that the performance of IWEB is very good when compared with other imbalanced data classification methods. An inventive cost-sensitive ensemble method for imbalanced data categorisation was presented by Tao et al. [146]. This approach combines a modified boosting algorithm with cost-sensitive Support Vector Machines. It also incorporates a self-adaptive sequential approach to determine misclassification cost weights. Their approach demonstrated superior generalisation performance in contrast to other existing techniques for classifying imbalanced datasets, as validated through extensive experimental results. For the purpose of tackling multi-class imbalance learning, Chen et al. [147] developed the double kernel-based class-specific BLS, which combines a double kernel mapping technique with class-specific penalty coefficients to improve performance on minority classes.

Cost-sensitive ensemble learning incorporates the concept of different misclassification costs for various classes into ensemble techniques. By parameter tuning of base classifiers to place larger misclassification costs on minority classes, this strategy addresses the issue of class imbalance by incentivising the model to pay more attention to these classes during training. This can lead to improved classification accuracy for minority classes, which is often the primary concern in imbalanced datasets. However, a potential disadvantage is that it requires careful tuning of the cost-sensitive parameters, parameter tuning increases the complexity of model learning and selecting appropriate cost values can be challenging. Meanwhile, it is often challenging to obtain accurate evaluation metrics when dealing with imbalanced data. Common metrics such as recall, accuracy, and F1 score might not adequately represent how well

the model performs on minority classes. Additionally, it may not always guarantee significant improvements, especially if the dataset is extremely imbalanced or if the choice of base learners or ensemble method is not well-suited to the problem.

5.4 | Hybrid Sampling for Imbalanced Data

Hybrid ensemble learning techniques have become effective resources for addressing the difficult issue of categorisation of unbalanced data. These methods combine the strengths of multiple algorithms and can be applied in a variety of ways to improve classification performance. One common approach is to ensemble resampling techniques within the ensemble framework, such as oversampling of minority class samples and undersampling of majority class samples. This helps to balance the class distribution during training. Another strategy is to combine the ensemble learning method with a powerful base classifier, which in turn is a completely new technique to solve the imbalance problem. By combining a number of strategies to increase the precision of minority and majority class classification, hybrid ensemble learning offers a flexible and efficient solution to the issue of imbalanced data classification.

In hybrid ensemble learning algorithms, on the one hand, hybrid sampling and ensemble learning techniques are combined to improve imbalanced data classification. For example, Yang et al. [148] provided a hybrid optimum ensemble classifier framework that combines useful methods using multi-objective optimisation with density-based under-sampling to effectively address the class imbalance issue. By integrating two data sampling techniques and two base classifiers with given weights to gain greater complementing benefits, Zhao et al. [149] devised a weighted hybrid ensemble approach for binary classification of unbalanced data. Xu et al. [150] developed an adaptive subspace optimisation ensemble approach for high-dimensional data imbalanced classification. It employs resampling to create class-balanced data for each base classifier and an adaptive subspace optimisation approach to create robust and discriminative subspaces.

On the other hand, ensemble learning also combines other techniques such as deep learning, active learning, and unsupervised techniques to address the issue of imbalanced data. For example, using histopathology pictures, In terms of execution time, number of parameters, and model size across multiple datasets, Garg and Singh [151] outperformed fine-tuned pre-trained models (ResNet50, InceptionV4, and MobileNetV2) with their ensemble model based on transfer learning for binary and multi-class breast cancer classification. This ensemble approach based on transfer learning can reduce the model training time, further enhancing learning efficiency. Chen et al. [135] introduced an ensemble learning approach integrated into deep CNNs to address class-imbalanced learning, using auxiliary classifiers from multiple hidden layers to focus on misclassified samples. Zhang et al. [152] proposed a reinforced online active ensemble learning approach equipped with enhancement mechanisms designed to handle concept drift and class imbalance in data streams, demonstrating superior classification performance and cost-effectiveness compared to semi-supervised and supervised algorithms across a variety of scenarios.

Hybrid ensemble learning methods for coping with imbalanced data classification offer several advantages, including enhanced model performance by combining multiple algorithms or techniques, improved robustness to class imbalance, and the ability to leverage the strengths of different approaches. They can effectively address the limitations of individual methods. However, hybrid ensembles can be computationally expensive due to the combination of multiple models, and they may require careful tuning of parameters. For example, incorporating deep learning techniques is a big problem in terms of computational time complexity, which may require long training times when dealing with large datasets, and in terms of incorporating active learning, which may need to cope with the performance inefficiencies associated with cold starts.

In summary, ensemble learning techniques offer a range of approaches to tackle imbalanced data problems. By combining multiple models and leveraging various sampling, pruning, and decision-making strategies, ensemble models can help address the challenges posed by imbalanced datasets and improve the predictive performance for minority classes.

6 | Ensemble Learning for High-Dimensional Data

Ensemble learning techniques have found valuable applications in handling high-dimensional data. High-dimensional data often contain numerous features, making traditional learning algorithms prone to overfitting, increased computational complexity, and reduced generalisation performance. Ensembles mitigate these issues by combining multiple base learners, each trained on different subsets of features or data, to create a more robust and accurate predictive model as shown in Figure 5. Techniques such as Random Forests, AdaBoost, and Bagging have been successfully applied to high-dimensional datasets, improving classification and regression tasks by capturing complex feature interactions, reducing the risk of overfitting, and enhancing predictive accuracy. Ensemble learning's ability to harness the collective intelligence of diverse models makes it a valuable tool for extracting meaningful

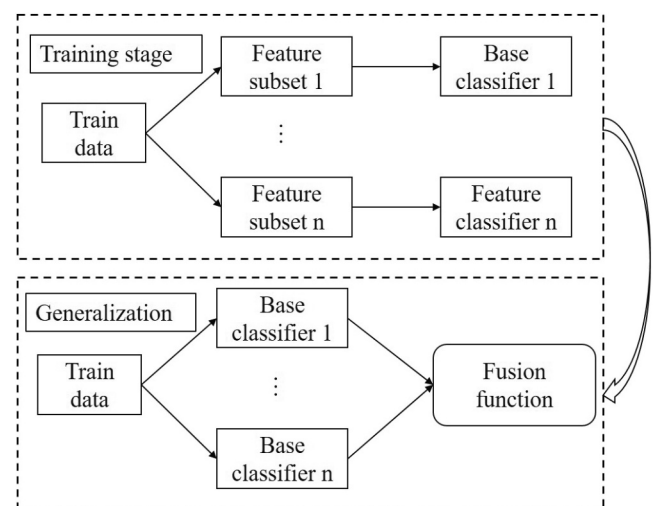


FIGURE 5 | Feature bagging training and generalisation process.

insights from complex, high-dimensional data, ultimately leading to better decision-making and problem-solving in various domains.

6.1 | Random Subspace for High-Dimensional Data

Random subspace and random forest methods select a random subset of features for each base model in the ensemble. By considering different subsets of features, ensemble models can capture diverse aspects of the data and reduce the risk of overfitting caused by irrelevant or redundant features.

When training machine learning models, a subset of randomly chosen characteristics or dimensions is taken into account. This technique is known as random subspaces, and it helps to minimise the ‘curse of dimensionality’ and improve the model’s generalisation performance by avoiding overfitting. For example, to build a set of closest neighbour classifiers in a high-dimensional feature space, Sampath et al. [153] contrasted random subspace and random projection approaches. When compared to typical nearest neighbour classifiers, both approaches perform better in prediction, according to experiments conducted on various datasets; which method is best for a given dataset will depend on its characteristics. Xu et al. [154] proposed an Improved Contraction-Expansion Subspace Ensemble (ICESE) method, which enhances high-dimensional imbalanced data classification by designing subspace selection and transformation optimisation strategies, employing a multilayer optimisation structure, and integrating a resampling scheme to construct a rebalanced subset for the base classifier. To deal with noisy datasets, Yu et al. [6] presented the Hybrid Incremental Ensemble Learning method, which considers both feature space and sample space, removing noisy attributes using bagging and linear discriminant analysis, incrementally selecting classifiers, and assigning weights based on specific criteria, which outperforms most of its ensemble counterparts on noisy real-world datasets.

In recent years, some researchers have combined semi-supervised learning with ensemble learning is also another innovative approach to subspace methods. Yu et al. [155] introduced a framework for incremental semi-supervised clustering that is specifically designed for high-dimensional data. This framework integrates a range of techniques, including random subspace methods, constraint propagation approaches, incremental set membership selection, and normalised cut algorithms. It significantly outperforms existing semi-supervised clustering ensemble methods on high-dimensional real-world datasets and successfully overcomes the drawbacks of conventional clustering ensemble methods. Yu et al. [156] presented a novel semi-supervised classifier ensemble framework (FSCE) designed to address the challenges of classifying high-dimensional data when the labelled training sample is limited. In this method, an auxiliary training set generation process, an adaptive feature selection process, and an adaptive weighting process are all included in the ASCE framework. Compared to most existing semi-supervised classifier integration techniques, ASCE has been shown to perform well, especially on high-dimensional datasets with less labelled training data.

Another generalised ensemble learning technique that is effective in handling high-dimensional data is random forests. It works by building several decision trees and combining their predictions to provide reliable and precise results. In high-dimensional data random forests, Ye et al. [157] presented a stratified sampling technique for feature subspace selection that successfully addresses the problem of choosing subspaces with few useful features and outperforms state-of-the-art methods in a variety of applications. For high-dimensional data in random forests, Wang et al. [158] presented a brand-new feature subspace selection technique dubbed PCA-SS. It improves performance in high-dimensional settings by enhancing feature selection inside decision trees using principal component analysis and stratified sampling.

Random subspace methods can improve ensemble performance by introducing diversity in the underlying classifier when dealing with high-dimensional data. However, there are some drawbacks to consider. Random subspace techniques have the potential to randomly choose a subset of features for every base classifier, hence omitting potentially important data. This may result in a loss of discriminative power. Furthermore, the performance of random subspace approaches could not be robust and dependable enough for real-world applications due to sensitivity to the size of the feature subset or the choice of random seeds.

6.2 | Bagging for High-Dimensional Data

The Bagging method has been effectively applied to high-dimensional data problems, such as feature selection and classification. The issue that Bagging is trying to solve with high-dimensional data is connected to the algorithm’s training stage. Bagging creates several random subspaces by choosing a particular number of characteristics from the original feature space to address the issues posed by high-dimensional data. These subspaces have their dimensions sampled randomly. On each of these subspaces, a corresponding number of base detectors, which are a class of classifiers, are trained. This approach helps mitigate the curse of dimensionality and overfitting issues commonly associated with high-dimensional data. To improve the overall prediction performance during the generalisation phase, a fusion function is used to merge the outputs of each of these basic detectors for each test point.

Bagging seeks to mitigate these issues by creating multiple subsets of the original dataset through a process called bootstrapping, where each subset contains a random selection of instances [18]. These subsets are used to train multiple base models independently. Pes [159] conducted an empirical study encompassing various feature selection algorithms and application domains to evaluate the robustness and utility of ensemble feature selection, finding that ensemble approaches tend to be more robust than single selectors across different classification tasks and feature subset sizes. Previous Bagging ensemble methods are usually developed based on variance reduction, which may deteriorate the performance of integration. Therefore, a number of researchers have proposed an ensemble detector that considers both variance and bias reduction [1.0–162]. For example, Wang et al. [160] introduced a

novelty detection for high-dimensional data that simultaneously considers both bias and variance reduction by categorising base detectors into different groups based on the dimensionality of their training subspaces, resulting in improved accuracy and diversity compared to traditional Feature Bagging and outperforming several state-of-the-art methods on various datasets. Wang et al. [161] introduced an ensemble classifier based on an improved adaptive network-based fuzzy inference system to provide compact fuzzy rule sets for processing high-dimensional data. This system uses improved bagging and dropout techniques. De et al. [163] extended semi-supervised ensemble methods to high-dimensional datasets, achieving better performance than a single model. For high-dimensional nonlinear dynamical systems, Chattopadhyay et al. [164] introduced a hybrid ensemble Kalman filter that makes use of a pre-trained deep learning-based data-driven surrogate. He et al. [162] presented Bayesian attribute bagging-based extreme learning machines, which calculate decision-making degrees of condition attributes using Bayesian decision theory, organise them into condition attribute groups, employs bagging attribute groups to train an ensemble of extreme learning machines, and uses information amount ratios as weights to combine predictions.

In conclusion, bagging techniques have a number of benefits when working with high-dimensional data. First, they enhance the robustness and stability of predictive models by reducing the variance associated with individual models. This is particularly beneficial in high-dimensional datasets where overfitting is a problem. Second, bagging can effectively deal with noisy data and outliers by averaging or voting across multiple models, which helps mitigate their impact on the final prediction. However, bagging has its drawbacks. It can be computationally expensive, especially when generating a large number of base models, which may be impractical for extremely high-dimensional datasets. In addition, while bagging reduces variance, it does not necessarily address bias, so if the underlying model is inherently biased, bagging may not provide a significant improvement.

6.3 | Regularisation Ensemble for High-Dimensional Data

Regularisation techniques, such as L1 or L2 regularisation (Ridge), can be incorporated into ensemble models to handle high-dimensional data. These regularisation methods encourage sparsity in the feature space, effectively selecting a subset of relevant features. Regularised ensemble models can help mitigate the curse of dimensionality by reducing the model's complexity and improving generalisation performance.

The difficulties of high-dimensional data analysis have been addressed in recent years by a number of frameworks and methods, such as regression using neural networks and least absolute shrinkage and selection operator (LASSO) models, dimensionality reduction-based algorithms, class noise detection, and combinations of ensemble models with resampling-based LASSO feature selection. For example, Cui and Wang [165] introduced a paradigm that combines prediction accuracy regularisation with model interpretation for high-dimensional

data regression. It involves building predictive models with neural networks that have random weights and creating ensemble features with LASSO models that have selected regularising variables. Jain and Xu [166] proposed a novel algorithm called RHDSI that effectively handles high-dimensional data with interaction effects, enabling classical statistical techniques to work on such data by integrating dimensionality reduction and machine learning for feature selection, demonstrating its superior or comparable performance to standard feature selection algorithms. Guan et al. [167] introduced a sequential ensemble noise filter (SENF) designed to address the challenge of detecting label errors in high-dimensional, mislabelled industrial informatics data by integrating feature selection and label error detection, demonstrating its superior performance compared to existing label noise detection methods on benchmark datasets. Patil and Kim [168] employed ensembles of regularised regression (ERRM) in conjunction with resampling-based LASSO feature selection (RLFS) to handle high-dimensional data with correlated features.

In summary, on the positive side, regularisation integration methods usually incorporate feature selection techniques such as LASSO to automatically identify relevant features and enhance the interpretability of the model. They also improve generalisation and reduce the risk of overfitting by combining multiple regularised models. In addition, they tend to produce more interpretable models that help understand the underlying factors driving predictions. However, these methods can be computationally intensive, especially for datasets with a large number of features, and may limit scalability. Parameter tuning can be complex and requires significant computational resources, and choosing the right combination of regularisation techniques and integration methods may require domain knowledge and experimentation.

6.4 | Ensemble Pruning for High-Dimensional Data

Ensemble pruning techniques can be applied to high-dimensional data to select a subset of base models that contribute most effectively to the ensemble's performance. Pruning can help reduce the complexity of the ensemble and remove base models that may be less relevant or redundant in the high-dimensional feature space.

In recent years, researchers have made significant strides in the field of high-dimensional data analysis by applying ensemble pruning techniques. These studies collectively highlight the growing importance of ensemble pruning techniques in tackling the challenges of high-dimensional data, ranging from clustering and regression to feature selection and classification tasks. Liu et al. [169] introduced a multi-objective evolutionary algorithm capable of optimising multiple clustering criteria concurrently without the need for predefined coefficients. They added a novel local search technique to the algorithm to improve convergence. To choose resilient solutions, they also developed a fuzzy ensemble method called knee-pruning. Through experimental evaluation across diverse datasets, their approach demonstrated effective handling of high-dimensional

clustering problems without the requirement for user-defined coefficients. Kim et al. [170] presented a filter-centric hybrid feature selection approach based on ensemble learning, aiming to balance computational efficiency with feature robustness and performance across a range of dataset types. Zhang et al. [171] proposed a framework for constructing an ensemble model of extreme learning machines for high-dimension data classification, using multiview-based random rotation ensemble pruning (MVRR-EP) to ensure diversity among component classifiers, resulting in improved classification performance on hyperspectral datasets. Xu et al. [24] introduced a multiview optimisation (MVO) approach to overcome the difficulties posed by high-dimensional class-imbalanced data, enhancing feature selection and diversity within an ensemble classifier.

Ensemble pruning methods offer a valuable approach to mitigating the complexities of high-dimensional data. They can significantly improve the efficiency and effectiveness of ensemble models by selectively retaining the most informative base learners while discarding redundant or noisy ones [24]. This leads to a more compact and computationally efficient ensemble, making it well-suited for handling high-dimensional data where resource constraints, such as memory and processing power, often come into play. However, the choice of base learners and the calibre of the pruning criteria have a major impact on how well ensemble pruning approaches work. The benefits of the technique can be undermined by suboptimal pruning caused by inaccurate criteria or poorly selected base learners.

When working with high-dimensional data, it is crucial to consider the computational resources required for training ensemble models. The increased number of features can significantly impact training time and memory requirements. Efficient implementation and optimisation techniques, such as parallelisation or approximation methods, may be necessary to handle high-dimensional ensemble learning effectively. In summary, ensemble learning techniques offer valuable strategies for handling high-dimensional data.

7 | The Applications of Ensemble Learning

As shown in Table 4, ensemble learning has emerged as a powerful technique across a diverse range of domains including security, finance, healthcare, transportation, recommender systems, computer vision, and so on [207]. By combining multiple complementary models, ensemble methods are able to improve predictive accuracy, robustness and generalisation compared to individual models. Key applications include intrusion and fraud detection, disease diagnosis, credit scoring, traffic flow prediction, personalised recommendations, image classification and data mining. This section will highlight a few typical application areas.

7.1 | Healthcare

The healthcare sector has given ensemble learning a lot of attention in the last few years [208]. Research has looked into

TABLE 4 | The applications of ensemble learning.

Domain	The application of ensemble learning
Healthcare	Medical imaging [172–175], bioinformatics [176, 177], healthcare systems [178–180]
Transportation	Traffic flow prediction [181, 182], route optimisation [183], passenger flow prediction [184–186]
Finance	Portfolio optimisation [187, 188], fraud detection [189, 190], algorithmic trading [191, 192]
Manufacture	Fault diagnosis [193, 194], production scheduling [195, 196], anomaly detection [197, 198]
Internet	Recommender system [199, 200], social network analysis [201, 202], intrusion detection [203–206]

the application of ensemble learning models to enhance performance and accuracy in a range of healthcare domains:

Based on an experiment conducted on 2933 lung CT images from COVID-19 patients, Tao et al. [172] discussed the use of deep ensemble learning models in the field of medical imaging to classify novel COVID-19 cases on CT images. This method may better satisfy the rapid detection requirements of the novel coronavirus disease COVID-19. An ensemble generative adversarial network system for abdominal image segmentation was proposed by Rezaei et al. [173], which addressed the issue of class imbalance and offered a more comprehensive generalisation than previous methods. Hossain et al. [175] presented a multi-class brain tumour detection model based on transfer learning and achieved improved classification accuracy and anomaly identification by leveraging average ensemble, outperforming the results obtained from a single model. According to Heisler et al. [174], optical coherence tomography angiography can detect diabetic retinopathy through deep ensemble learning. Their proposed methods offer the potential for clinically valuable solutions in diabetic retinopathy diagnoses.

In the field of bioinformatics, Guo et al. [176] proposed ensemble embedded feature selection for multi-label bioinformatics data classification. Using sequence-based data, Tang et al. [177] suggested ensemble learning methodologies for the prediction of bacterial small RNAs. They created neural network ensembles and weighted average ensembles to increase the precision of small RNA prediction.

In the field of healthcare systems, a smart diabetes recommendation system is proposed by Ihnaini et al. [178]. Applying advanced ensemble learning, the system makes intelligent decisions for diabetes patients. A clinical decision support system was developed by De et al. [180] using the dependable semi-supervised ensemble learning technique. According to their research, this method of predicting urinary tract infections is more accurate than urine cultures or urinalysis. Kumar et al. [179] discussed ensemble learning in the context of Healthcare 4.0, which integrates industry 4.0 concepts into healthcare and

improves the performance of blockchain-based decentralised applications for the smart healthcare system. They explore the use of optimisation algorithms and blockchain-based decentralised applications to enhance the performance of smart healthcare systems.

Overall, ensemble learning has been utilised to enhance disease detection, optimise therapy recommendations, and boost system performance across a variety of healthcare disciplines.

7.2 | Transportation

Various aspects of transport have been improved with ensemble learning, such as traffic flow prediction, route optimisation, and passenger flow prediction. Wang et al. [182] proposed a short-time traffic flow prediction model considering spatio-temporal factors, integrating multiple LSTM models to improve the accuracy and stability of prediction. XGBoost and LSTM neural networks were combined into an ensemble model for short-term traffic flow predictions by Liu et al. [181]. Wang et al. [184] applied a weighted ensemble approach combining ARIMA, SVR, and GBDT models for passenger flow forecasting in metro systems. Feng et al. [183] constructed a multi-objective optimisation model for the dynamic vehicle routing problem with time window and combined ensemble methods to enhance the algorithm's performance. A hybrid model combining gated recurrent unit and ensemble empirical mode decomposition was presented by Liu et al. [185], yielding better and faster results for short-term passenger flow prediction. Li et al. [186] developed a novel approach for multi-source time series fusion and direct interval prediction based on deep ensemble learning, effectively capturing the dynamic patterns of metro passenger flow.

Overall, studies have shown combining diverse base learner models using ensemble techniques can enhance predictive performance on key intelligent transportation system tasks. However, challenges persist in aspects such as the complex data preprocessing of transportation systems, model interpretability, and real-time performance. To better harness the role of ensemble learning in transportation systems, future research needs to further address these challenges and seek effective solutions.

7.3 | Finance

There are many applications of ensemble learning in other areas of finance as well, such as portfolio optimisation, risk modelling, fraud detection, algorithmic trading, and time series forecasting, among others. To mitigate estimation error in small samples, Shen et al. [188] developed an ensemble learning growth optimal portfolio strategy combining bootstrap aggregation with random subspace. In order to estimate the future return of assets, Zhou et al. [187] combined six separate prediction models utilising the ensemble learning approach, whereby the assets with greater prospective returns are picked for portfolio optimisation. By comparing the effectiveness of five ensemble learning models based on bagging and boosting, Chen et al. [189] clarified the mechanism by which ensemble learning affects financial fraud detection and expanded relevant research

in the financial sphere. The dynamic ensemble selection approach, which dynamically combines various classifiers to get a final prediction, was first presented to the fraud detection literature by Achakzaiz et al. [190]. Hao et al. [191] utilised reinforcement learning and a three-dimensional fuzzy vector to analyse current stock trends and other market indicators, applying an average ensemble method to dictate trading actions, which showcased enhanced performance compared to two benchmark methods. Lahmirmi et al. [192] advanced ensemble learning models and classification systems for financial data classification, finding AdaBoost to be particularly effective and ensemble systems to generally outperform existing models in terms of reliability and consistency.

In summary, properly designed ensemble learning architectures have been shown to enhance decisions, reduce risk, detect fraud, exploit signals, and improve predictive accuracy in finance. The flexibility of ensembles allows financial institutions to build tailored solutions by combining diverse machine learning models.

7.4 | Manufacture

Ensemble learning in the manufacturing industry has garnered widespread attention [15, 209, 210]. Its application in manufacturing can optimise fault diagnosis, production scheduling, anomaly detection etc, thereby enhancing production efficiency, product quality, and safety. Firstly, ensemble learning plays a crucial role in fault detection within manufacturing. An intelligent ensemble approach that integrates many members is designed by Zhang et al. [193], offering a possible substitute for defect diagnostics in refrigeration systems. Li et al. [194] proposed a domain-adaptive method for fault diagnosis, employing adversarial training within the ensemble to achieve effective results, especially in diagnosing faults with domain mismatch. Simultaneously, the application of ensemble learning in production scheduling contributes to increased production efficiency. Gil et al. [195] integrated various evolutionary algorithms to meet online scheduling requirements. ***strokumić et al. [196] utilised ensemble learning based on priority rules to enhance task scheduling performance under limited resources. Moreover, ensemble learning can be applied to anomaly detection to enhance product quality in manufacturing. Alelaumi et al. [197] used AdaBoost to detect anomalies in the stencil printing process, improving the effectiveness and reliability of anomaly detection in production processes and reducing reworking costs for large-scale production. Chan et al. [198] employed deep ensemble learning to address anomaly detection in multi-class systems, achieving detection of unknown classes.

In conclusion, the application of ensemble learning in the manufacturing industry holds extensive prospects and potential. With the ongoing digital transformation and intelligent upgrades in manufacturing, the application of ensemble learning is expected to further develop and strengthen. It is essential to continually explore and innovate the application patterns and methods of ensemble learning in manufacturing to adapt to evolving market demands and technological developments.

7.5 | Internet

Ensemble learning has made substantial progress in a number of online domain areas, including intrusion detection, social network analysis, and recommendation systems [211], thanks to the Internet's rapid expansion. In the realm of recommendation systems, ensemble learning techniques can enhance the accuracy and diversity by combining algorithms such as content-based recommendation, collaborative filtering, and deep learning recommendation. Shankar et al. [199] presented an intelligent recommendation system, utilising ensemble learning to reduce duplicate and irrelevant recommendations for customers. A recommendation model for determining user preferences was designed by Choudhary et al. [200] using a deep ensemble learning technique. In the domain of social networks, ensemble learning provides powerful analytical tools. A multi-view ensemble learning-based emotion role-mining technique was presented by Du et al. [201] to analyse network emotion contagion. Additionally, Bibi et al. [202] introduced an unsupervised learning framework based on concepts and hierarchical clustering for Twitter sentiment analysis, achieving experiment results comparable to supervised learning methods. Furthermore, ensemble learning has found widespread applications in intrusion detection. Vo et al. [203] combined ensemble learning and generative adversarial networks for real-time intrusion detection. Gadekallu et al. [204] proposed an ensemble machine learning model based on Moth-Flame Optimisation for intelligent traffic system intrusion detection. Singh et al. [205] designed an intrusion detection model for cloud computing networks using ensemble learning methods, achieving higher detection rates and minimal false alerts compared to state-of-the-art methods. Rajadurai and Devi Gandhi [206] utilised stacked ensemble learning to construct an intrusion detection system, demonstrating superior accuracy compared to other intrusion detection models.

In summary, the application of ensemble learning in the Internet domain holds extensive prospects and potential. It can leverage diverse algorithms and technologies to enhance the performance and accuracy of various Internet applications. With continuous technological advancements and growing data, the application of ensemble learning in the Internet domain is poised for further expansion and development.

8 | Challenges and Future Directions

Ensemble learning has undeniably achieved remarkable success. However, it still faces several challenges and offers promising future directions for improvement and applicability. Researchers can explore various avenues to address these challenges:

- **Enhancing Diversity:** A crucial challenge in ensemble learning is ensuring diversity among individual models. To avoid redundancy and improve overall performance, future research can focus on techniques that encourage diversity. This might involve using different learning algorithms, diverse training data subsets, or introducing randomness in the learning process.

- **Scalability:** Ensemble learning is known to be computationally expensive as the number of models increases. Taking care of big ensembles can be tough because much memory is used for training, and predictions take a while. Future research should explore efficient algorithms and distributed computing techniques to scale ensemble learning for big data scenarios.
- **Automating Model Selection and Combination:** Selecting the best subset of models from a large ensemble and determining optimal combination weights can be challenging. Future directions can include investigating advanced techniques, such as genetic algorithms, reinforcement learning, or Bayesian optimisation to automate the ensemble selection and combination process.
- **Advanced Combination Strategies:** Combining the predictions of individual models is a critical step in ensemble learning. Traditional combination methods may not fully exploit the ensemble's potential. Future research can explore advanced combination strategies to improve prediction fusion and enhance ensemble performance.
- **Adaptivity to Concept Drift:** Many real-world applications involve dynamic data environments with concept drift. Future research should focus on developing online and adaptive ensemble learning methods that can dynamically update the ensemble to accommodate changing data distributions and conditions.
- **Interpretability and Trustworthiness:** Ensemble learning is often considered a 'black-box' model due to its challenging interpretability of decision processes, which poses a problem for applications that require high interpretability and trustworthiness. Enhancing the interpretability and trustworthiness of ensemble models is a crucial research direction in ensemble learning.
- **Real-time Applications:** Ensemble learning typically exhibits higher time complexity compared to single models, limiting its applicability in applications requiring real-time responsiveness. Additionally, effectively training and deploying ensemble models on large-scale datasets pose significant challenges.

9 | Conclusion

Overall, ensemble learning is a valuable technique in machine learning that has been widely adopted due to its ability to address complex learning problems. Its effectiveness in enhancing predictive performance, reducing overfitting, and improving generalisation across diverse datasets has been widely acknowledged. It continues to be an active area of research and development, with ongoing efforts to develop new ensemble methods and optimise their performance for different applications. In this paper, a four-layer research framework is established for the research of ensemble learning, which can offer a comprehensive and structured review of ensemble learning from bottom to top. The review first discusses recent advancements in ensemble learning, including novel ensemble techniques, such as deep ensemble learning, semi-supervised ensemble learning, and so on. Then, it explores different

ensemble learning approaches to handle challenging datasets, such as imbalanced datasets, high dimensional datasets, and so on, in different domains. Finally, the review also identifies current challenges and outlines potential future directions for ensemble learning. In summary, this review aims to provide researchers and practitioners with a comprehensive overview of ensemble learning methods, to elucidate the strengths and weaknesses of these methods, and to provide guidance for their sensible application in real-world scenarios.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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